

Project Story

“One of my recent projects is an Air Quality Forecasting and Time-Series Analysis project using Python.

I worked with around **9,357 time-series observations** containing environmental variables such as temperature, relative humidity and multiple air-quality sensor measurements.

My main focus was Relative Humidity. I processed the timestamp column using Pandas and analyzed forecast values represented by `yhat`. I created interactive Plotly visualizations to show the forecast over time, compare actual RH with forecasted RH, and visualize the forecast uncertainty using `yhat_lower` and `yhat_upper`.

One important part of the project was data-quality validation. I discovered that the RH column contained **366 observations with a value of -200**, which represents an invalid or missing sensor reading. Therefore, I excluded those observations when evaluating the forecast.

After cleaning those invalid RH values, I calculated additional evaluation metrics. The forecast achieved an **MAE of 9.42, RMSE of 12.14 and an actual-versus-forecast correlation of approximately 0.714.**

The forecast captured the overall temporal pattern, although the actual RH values showed significantly more short-term variation.

The biggest lesson from this project was that forecasting is not just about generating a prediction. **Data quality, uncertainty intervals and objective evaluation are equally important.**

I used Python, Pandas, NumPy, Matplotlib, Seaborn and Plotly throughout the project.”**